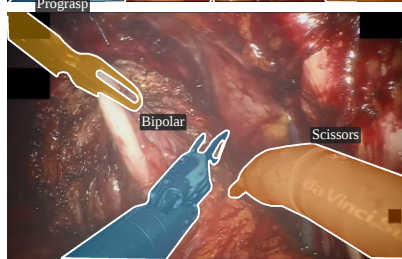
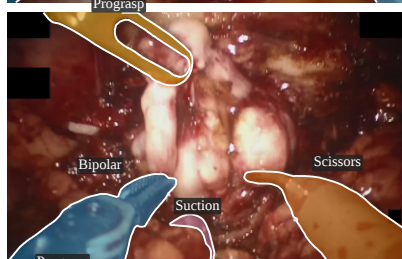
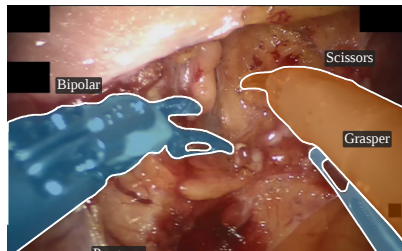
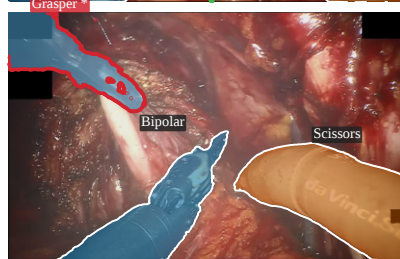
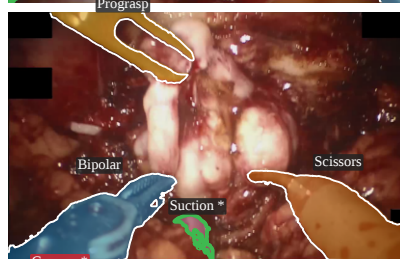
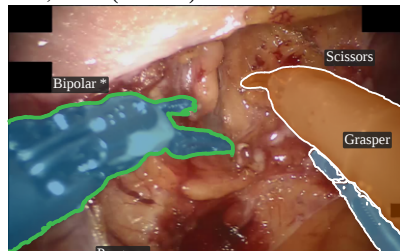


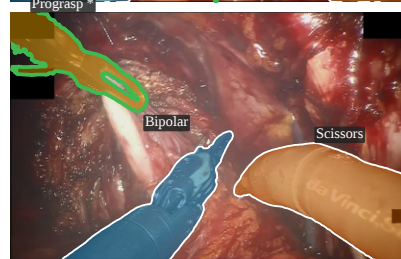
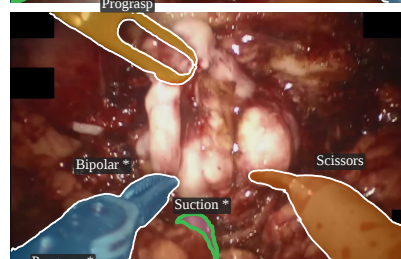
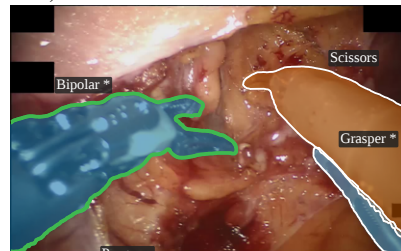
Ground truth



Ours, causal (real time)



Ours, non-causal



■ Bipolar
 ■ Prograsp
 ■ Needle driver
 ■ Scissors
 ■ Suction
 ■ Clip applicator
 ■ Grasper

* sent to tracking green outline: fixed by tracking red outline: wrong class